Fiscal Year:	FY 2006	**Task Last Updated:**	FY 09/30/2005
PI Name:	Freed, Lisa E M.D., Ph.D.		
Project Title:	Microgravity Tissue Engineering		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:	**TechPort:**		No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	02139-4037	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-08
Start Date:	12/01/2003	**End Date:**	11/30/2007
No. of Post Docs:	8	**No. of PhD Degrees:**	1
No. of PhD Candidates:	8	**No. of Master' Degrees:**	0
No. of Master's Candidates:	6	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Changed Division and Discipline/Program to FSB as of FY2006, per program changes at that time, per J. Cohen/HQ and JSC/J. Dardano and previous information fr D. Risin/JSC in Mar 2006 (jvp 6/2009) NOTE: Funding reinstated to 11/30/2007, per J. Cohen (HQ, 8/2006)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Vunjak-Novakovic, Gordana (Columbia University and M.I.T.) Rupnick, Maria (Harvard Medical School) Niklason, Laura (Duke University Medical Center) Guilak, Farshid (Duke University Medical Center) Padera, Robert (Harvard Medical School)		
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Performance Goal No.:	1.21		
Performance Goal Text:	Sustain a leading biotechnology research program that will assure continued scientific and technical leadership.		

Task Description:

Our research project involves cell- and tissue-levels studies of 3D tissue constructs using models of relevance to tissue loss in astronauts in microgravity and skeletal and myocardial tissue repair in human patients on earth. We selected to study skeletal and cardiac tissues because: (a) spaceflight adversely affects these tissue types, (b) improved treatments are needed for millions of patients who suffer from progressive skeletal and cardiac diseases, (c) these two distinctly different tissue types can provide paradigms for studies of individual and interactive effects of biophysical stimuli, and (d) we and others have already demonstrated the feasibility of engineering skeletal and cardiac-like tissue, and thus can capitalize on recent advances to promote groundbreaking research and technology transfer.
We previously designed an in vitro system based on 3D cell-biomaterial constructs and advanced bioreactor systems, and demonstrated that bioreactor-grown constructs were larger and functionally superior to controls grown in petri dishes, as assessed by quantitative structural, molecular, and functional assays. The current limitations of our tissue engineering research are as follows: (i) functional cartilage constructs based on human cells have not yet been made, and incomplete integration of cartilage constructs with adjacent host tissues may limit their long-term mechanical properties, (ii) cardiac tissue constructs have only been made in very thin layers and from non-human cells that are present at low number and low viability, and (iii) spaceflight tissue culture studies are limited by the lack of proper controls necessary to derive underlying mechanisms. We are addressing these limitations by: (a) integrated studies in cell and molecular biology, mass and momentum transport, design and operation of bioreactors including microgravity analog systems, tissue physiology and biomechanics, and implant biocompatibility and remodeling, (b) experiments interpreted with the aid of mathematical models, (c) complementary in vitro and in vivo studies, and (d) functional evaluation at various hierarchical scales, i.e. molecular, cell, and tissue.

The project is organized as two Specific Aims: (1) functional human skeletal tissue engineering, and (2) functional cardiac tissue engineering. The global hypothesis is that biophysical stimuli that are physiologic in nature, magnitude and regime of application can direct cells to regenerate functional tissues in vitro. We can test this hypothesis in controlled studies of cells cultured on 3D scaffolds in bioreactors and in the presence of biophysical stimuli. The work is expected to help define how biophysical cues regulate skeletal and cardiac tissue development and function, and to yield engineered skeletal and cardiac tissues with superior functionality to that previously reported in the literature. Our studies of skeletal and cardiac tissue constructs involve the cultivation of cells on 3D scaffolds in bioreactors, including microgravity analog systems, and in the presence of biophysical stimuli. Critical research areas include cell sourcing (cell expansion, stem cell differentiation), development of novel biomaterial scaffolds, cell seeding of scaffolds, construct cultivation in vitro for controlled studies of biophysical stimuli (e.g. hydrodynamic shear, cyclic stretch, growth factors, pharmaceutical agents), and construct implantation in vivo for studies of construct survival, integration, and capacity for defect repair. The constructs are assessed with respect to structure, composition, and mechanical function to evaluate individual and interactive effects of experimental variables. Key equipment includes facilities for working with cells, explanted tissues, and experimental animals, polymer processing equipment, bioreactor systems, biomechanical and electrophysiological test systems, and analytical instruments for histology, biochemistry, molecular biology, microscopy and image analysis. Construct properties are correlated with in vitro culture conditions and the outcome of in vivo studies, in order to provide feedback necessary to tailor model system parameters and obtain the best overall result. Our research is relevant to Critical Path Roadmap (CPR) questions #2) impaired fracture healing, #3) injury to joints, #5) occurrence of serious cardiac dysrhythmia, #6) diminished cardiac function, #19) pharmacology of space medicine, and possibly also #21) rehabilitation on Mars < http://bioastroroadmap.nasa.gov/ ;. Our combined use of 3D tissue constructs and advanced bioreactor systems provides a unique model system for controlled in vitro studies at the cell and tissue-levels that may be relevant to treating skeletal and myocardial tissue loss and dysfunctions in astronauts. Biophysical stimuli play a critical role in tissue growth, maintenance, and remodeling. During human spaceflight, altered biophysical stimuli and the resulting systemic adaptations can cause fluid and electrolyte imbalance, altered neurohormonal regulation, diminished tissue remodeling, and loss of bone and muscle mass. Exercise regimens and pharmaceutical interventions have been proposed to try to maintain skeletal and cardiac tissues in-flight, and to try to rehabilitate deconditioned skeletal and cardiac tissues post-flight. However, the ability to determine the underlying mechanism(s) or assess the potential risks/benefits of currently used countermeasures is limited by a lack of experimental models in which to study development and remodeling of skeletal and cardiac tissues. Our controlled studies, in high fidelity in vitro models, of how 3D constructs respond to biophysical stimuli are expected to help identify how the microgravity environment of space may compromise the ability of skeletal and cardiac tissues to respond to and/or withstand biophysical stimuli. Improved understanding is expected to aid in the development of better countermeasures.

Our tissue engineering research has implications not only for astronauts but also for humans on earth, e.g. for obesity related degenerative joint diseases and hypertensive cardiomyopathy. Skeletal and cardiac tissues perform structural and mechanical functions vital for health and survival. Incidences of osteoarthritis and heart failure are increasing and, in the absence of curative interventions, have resulted in substantial human suffering and medical expense. Over forty million Americans suffer from arthritis, which is characterized by the degeneration of cartilage and results in chronic pain and progressive disability. Once damaged, cartilage has a limited capacity for self-repair, and no currently available therapy can predictably restore a functional articular surface. Over five million Americans suffer from heart failure, which is characterized by the degeneration of cardiac muscle and is a leading cause of morbidity and mortality in developed countries. The only currently available definitive treatment for end stage heart failure is cardiac transplantation, but limited organ availability leads to long waiting periods that are often not survivable. Tissue engineering is emerging as a significant clinical option to address tissue and organ failure by implanting a 3D tissue construct that can provide immediate functionality and the capacity to grow and integrate with surrounding host tissues. However, although cell-based procedures for articular cartilage repair procedures are already in widespread clinical use, critics maintain that the field is principally empirical in nature, and the results unsatisfactory with respect to integrative repair and durability. Significant challenges also remain to be solved before cell-based procedures for cardiac repair can be used clinically. Our research is thus expected to be of critical importance for the human exploration of space and also to benefit life on Earth.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:	Skeletal and cardiac tissues perform structural and mechanical functions vital for health and survival. Incidences of osteoarthritis and heart failure are increasing and, in the absence of curative interventions, have resulted in substantial human suffering and medical expense. Over forty million Americans suffer from arthritis, which is characterized by the degeneration of cartilage and results in chronic pain and progressive disability. Once damaged, cartilage has a limited capacity for self-repair, and no currently available therapy can predictably restore a functional articular surface. Over five million Americans suffer from heart failure, which is characterized by the degeneration of cardiac muscle and is a leading cause of morbidity and mortality in developed countries. The only currently available definitive treatment for end stage heart failure is cardiac transplantation, but limited organ availability leads to long waiting periods that are often not survivable. Tissue engineering is emerging as a significant clinical option to address tissue and organ failure by implanting a cell-based tissue substitute that can provide immediate functionality and the capacity to grow and integrate with surrounding host tissues. However, although cell-based procedures for articular cartilage repair procedures are already in widespread clinical use, critics maintain that the field is principally empirical in nature, and the results unsatisfactory with respect to integrative repair and durability. Significant challenges also remain to be solved before cell-based procedures for cardiac repair can be of clinical use, especially in the areas of cell sourcing and functional construct integration. Our research is thus expected to yield demonstrable benefits on earth.
Task Progress:	Aim 1 (selected progress). Our skeletal tissue engineering studies involve chondrocytes or chondroprogenitor cells, 3D scaffolds, bioreactors, and biophysical signals. In complementary studies (Tognana, Osteoarthritis Cartilage, 2005a; Tognana, Osteoarthritis Cartilage, 2005b), we explored the development and remodeling composites made of cartilage constructs and explants of native cartilage or bone. Studies were done in vitro in rotating bioreactors and in vivo in a rodent model (ectopic implants in nude mice) to provide more controllable conditions and lower cost than other models (orthotopic implants in large animals). In vitro, constructs based on chondrocytes expanded in monolayers deteriorated over time, underscoring the need to maintain differentiated phenotype during cell expansion. Scaffold degradation rate was also a critical variable, the appropriate selection of which depended on cell type and culture conditions. In vivo, the presence of adjacent cartilage inhibited development of the cartilage construct. In contrast, the presence of adjacent bone stimulated repair responses (e.g. a wide cartilaginous zone comprising chondrocytes aligned in columns and trabecular rebuttressing by new bone formation) that were similar in some respects to processes that occur during the healing of fractured bones. These data add to a growing body of literature addressing remodeling of cartilage and bone. In another study (Marolt, Trans Orthop Res Soc, in review), we tested whether rotating bioreactors can be used in conjunction with human bone marrow stromal cells and silk sponges to generate bone- and cartilage-like constructs. Constructs cultured in media supplemented with either osteogenic or chondrogenic growth factors were assessed by a variety of methods including imaging by microCT and contrast-enhanced MRI. Bioreactor culture of constructs in osteogenic media yielded large bone-like constructs in which the volume fraction of mineralized tissue was comparable to that in human lumbar vertebral bone. The positive outcome for bone was attributed to combined effects of osteogenic growth factors and high mass transfer rates in rotating bioreactors. In contrast, chondrogenic media induced very little cartilage development in vitro, possibly due to a low cell density or low chondrogenic potential of the initial cell population. Interestingly, low concentrations of glycosaminoglycans, a key cartilage component, could be visualized by MRI but not by histology. The data show that rotating bioreactors provide a good model system for generating large bone-like constructs from human cells. We also developed a novel scaffold for cartilage tissue engineering that mimics some of the complex material properties (i.e. anisotropy and tension-compression nonlinearity) of native articular cartilage (Moutos, Trans Orthop Res Soc, in review). A computer-controlled loom was used to weave a 3-dimensional fabric from polyglycolic acid yarns. Initial scaffolds displayed approximately three orders of magnitude difference in tensile and compressive moduli, and tensile anisotropy in the x-y plane. Scaffolds seeded with chondrocytes in fibrin gel and cultured for 1 month in vitro had significantly higher compressive moduli than control unseeded scaffolds. These data show that 3D weaving technologies can be used to make scaffolds suitable for cell culture that mimic some of the complex biomechanical properties of native articular cartilage. Aim 2 (selected progress). Our cardiac tissue engineering studies involve heart cells, 3D scaffolds, bioreactors, and biophysical signals. In one study (Radisic, PNAS USA, 2004), a collagen sponge was seeded with cardiomyocytes in Matrigel and cultured in vitro in the presence of electrical field stimulation. Electrical stimulation induced cell elongation and alignment to yield constructs with remarkable ultrastructural organization and mechanical contractility, suggesting for the first time that electrical signals can have beneficial effects on the functional assembly of cardiac-like tissue. In another study (Radisic, Tissue Eng, in review), an elastomeric sponge made of polyglycerol sebacate was punched to produce an array of channels, seeded with cardiomyocytes in Matrigel, and perfused with media. Constructs cultured in media supplementated with perfluorocarbon particles, which provided a reservoir of dissolved oxygen had higher amounts of DNA and muscle markers and better contractile properties than controls cultured in unsupplemented media. A mathematical model of oxygen distribution was developed (Radisic, AJP-Heart, 2005) and oxygen gradients were correlated with construct cell density and viability (Radisic, Biotech Bioeng, in review). We also evaluated novel cardiac tissue engineering scaffolds in an attempt to improve construct mechanical properties, as previously tested materials were either not elastic (i.e. polyglycolic acid fibers) or very weak (i.e. collagen sponges). In one study (Boublik, Tissue Eng, 2005) an elastomeric knitted fabric made of benzylated hyaluronan was seeded with cardiomyocytes in a fibrin gel and cultured in vitro. Initially, ultimate tensile strength and failure strain were determined by the fabric and were significantly higher than native heart, whereas construct stiffness was mainly determined by the fibrin and was half as high as native heart. Over 1 week of culture, cell-mediated, serum-dependent remodeling was demonstrated by an increase in construct collagen content and maintenance of stiffness not observed in cell-free constructs. Cyclic stretch further increased construct collagen in a manner dependent on loading regime. However, although the construct mechanical properties might suffice for cardiac repair applications, macroscopic contractions could not be observed and the component cells remained round and immature, suggesting that the combination of knitted fabric and fibrin and did not optimally support cardiogenesis. In another study (Park, In Vitro Cell Dev Biol, in press) an elastomeric sponge made of polylactide-co-caprolactone was coated with collagen, seeded with cardiomyocytes in Matrigel and cultured in vitro. The resulting constructs contracted spontaneously and in response to electrical stimuli, and exhibited better properties than control constructs based on elastomers of lower (polylactide-co-glycolide) or higher (collagen) compliance. Together, these studies suggest that elastomeric properties of the scaffold are a key determinant of electromechanical signal propagation in cardiac constructs. We also evaluated whether 3D tissue constructs could be used as a high fidelity experimental model for studies of pharmaceutical agents. In one study (Pedrotty, AJP-Heart, 2005), a non-woven mesh made of polyglycolic acid was

	seeded with skeletal myoblasts and cultured in vitro in the presence of electrical stimulation and/or cardiomyocyte-conditioned culture media. Electrical stimulation induced myoblast proliferation in a manner that was selectively ablated by adding Verapamil, a calcium channel blocker. In other studies (Cheng, unpublished), sponges made of collagen were seeded with cardiomyocytes and Matrigel and cultured in media supplemented or not with Digoxin, a positive inotropic agent. Preliminary findings indicated that Digoxin had dose-dependent affects on cell metabolic activity (based on the molar ratio of lactate to glucose in media), construct molecular composition (based on amounts of muscle markers by semi-quantitative Western blot), and macroscopic electrophysiological function (excitation threshold and contractile amplitude). Together, these studies add to a growing body of evidence that 3D tissue constructs can provide a high-fidelity model to for molecular-, cell- and tissue-level studies of clinically used cardiovascular drugs.
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